Ogbonna Arinze

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SUMMARY

Frontend Engineer with 8 years of experience architecting and shipping complex, data-driven web applications using React and TypeScript across healthcare, fintech, and civic tech. Specialized in building rich interactive interfaces, including data visualization dashboards, design systems with WCAG accessibility, and performance-critical UIs handling real-time streaming and large-scale datasets. Proven technical leadership in mediating architectural decisions, maintaining component libraries, and mentoring engineers. Experienced contributor to open-source tooling with a track record of delivering semantically versioned, well-documented libraries.

CORE COMPETENCIES

- **Complex UI Architecture:** Rich interactive interfaces, data-driven UIs, design systems, scalable component architectures, monorepos (Nx, Turborepo), and SSR/SSG.
- **Data Visualization:** Recharts (22 chart types), react-chartjs, real-time streaming dashboards, and interactive graph-based interfaces.
- **Core Technologies:** React, TypeScript, Next.js, modern HTML5 & CSS3, Tailwind CSS, Radix UI, Redux, Zustand, and React Query.
- **Accessibility & Standards:** WCAG 2.2, ARIA patterns, semantic HTML, responsive design, and pragmatic accessibility techniques.
- **Performance Optimization:** Code splitting, lazy loading, CDN strategies, Lighthouse, Chrome DevTools, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer), and Core Web Vitals.
- **Testing & Quality:** Playwright, Jest, Cypress, Storybook, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Technical Leadership:** Architecture reviews, standards mediation, mentorship, semantic versioning, library documentation, and Agile/Scrum.
- **Infrastructure:** AWS (CloudFront, S3), Vercel, Docker, GitHub Actions, CI/CD pipelines, and Nx monorepo orchestration.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, establishing consistent component APIs, **WCAG-compliant** accessibility patterns, and semantic versioning across multiple product teams, improving UI consistency and developer velocity.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension with **TypeScript**, architecting complex data-driven interfaces for clinical site integrations and EHR system connectivity.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using OpenAI GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, building the rich conversational UI with real-time streaming responses and embeddable widget architecture.
- Optimized rendering performance for complex interactions by implementing **lazy loading, code splitting, dynamic imports**, memoization to minimize re-renders, and **CDN asset delivery**, measuring improvements via **Lighthouse** and **Chrome DevTools** profiling.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components with hooks, streamlining data flow and improving maintainability across the application.
- Implemented **JWT-based authentication** and **role-based access control** in the React frontend using HTTP-only cookies for secure token handling and permission-based UI rendering, collaborating with backend engineers on auth strategy alignment.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Provided **technical leadership** by mediating discussions on implementation standards, conducting architecture reviews, and mentoring a team of two junior developers on React patterns and code quality practices.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of a complex **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend rendering performance by profiling bottlenecks with **React Profiler** and **Lighthouse Audits**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time** across devices.

- Drove a **40% improvement** in application performance by optimizing caching mechanisms, significantly boosting speed and responsiveness for data-heavy views.
- Utilized **Storybook** for documenting React components with well-structured, semantically organized component documentation, streamlining change management and onboarding for the engineering team.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and more reliable feature delivery.
- Established engineering standards for code quality using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, conducting code reviews and fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built **dynamic data visualization dashboards** using **React** and **react-chartjs**, rendering complex financial datasets with interactive charts for transaction monitoring, settlement tracking, and reconciliation reporting.
- Built a high-volume bank **Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%** through complex, data-rich UI workflows.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API integration.
- Adopted the native browser **EventSource API** for real-time data streaming, replacing conventional socket.io methods to deliver faster updates and a more responsive interface for live transaction monitoring.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent and intuitive user interface across the application.
- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable code.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Developed **SureEdu**, a school management portal in a team of 5, building interactive **JavaScript** frontend interfaces for staff payroll tracking, student records management, and guardian communications.
- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Devised comprehensive documentation for each application, detailing all functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered civic data platform with complex data visualizations, real-time streaming, and interactive graph-based interfaces.

- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring **22 interactive chart types** auto-generated from AI responses, including bar, line, area, pie, radar, treemap, and funnel visualizations with dark mode and responsive mobile layouts.
- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, Next.js citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, URL state management with nuqs, shareable public conversations with **SEO optimization** (Open Graph metadata, JSON-LD FAQPage schema, slug-based URLs with SSR), and responsive mobile-first layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*